University of Applied Sciences

HOCHSCHULE EMDEN·LEER

# Machine Learning

## Knowledge Discovery in Databases Process

Prof. Dr. E. Wings

# Agenda

1. Data Mining - Procedure models
2. Knowledge Discovery in Databases Process
3. Modeling Process
4. Database
5. Selection
6. Data Preparation
7. Data Transformation
8. Data Mining
9. Model Patterns
10. Evaluation, Verification

# Data Mining - Procedure models

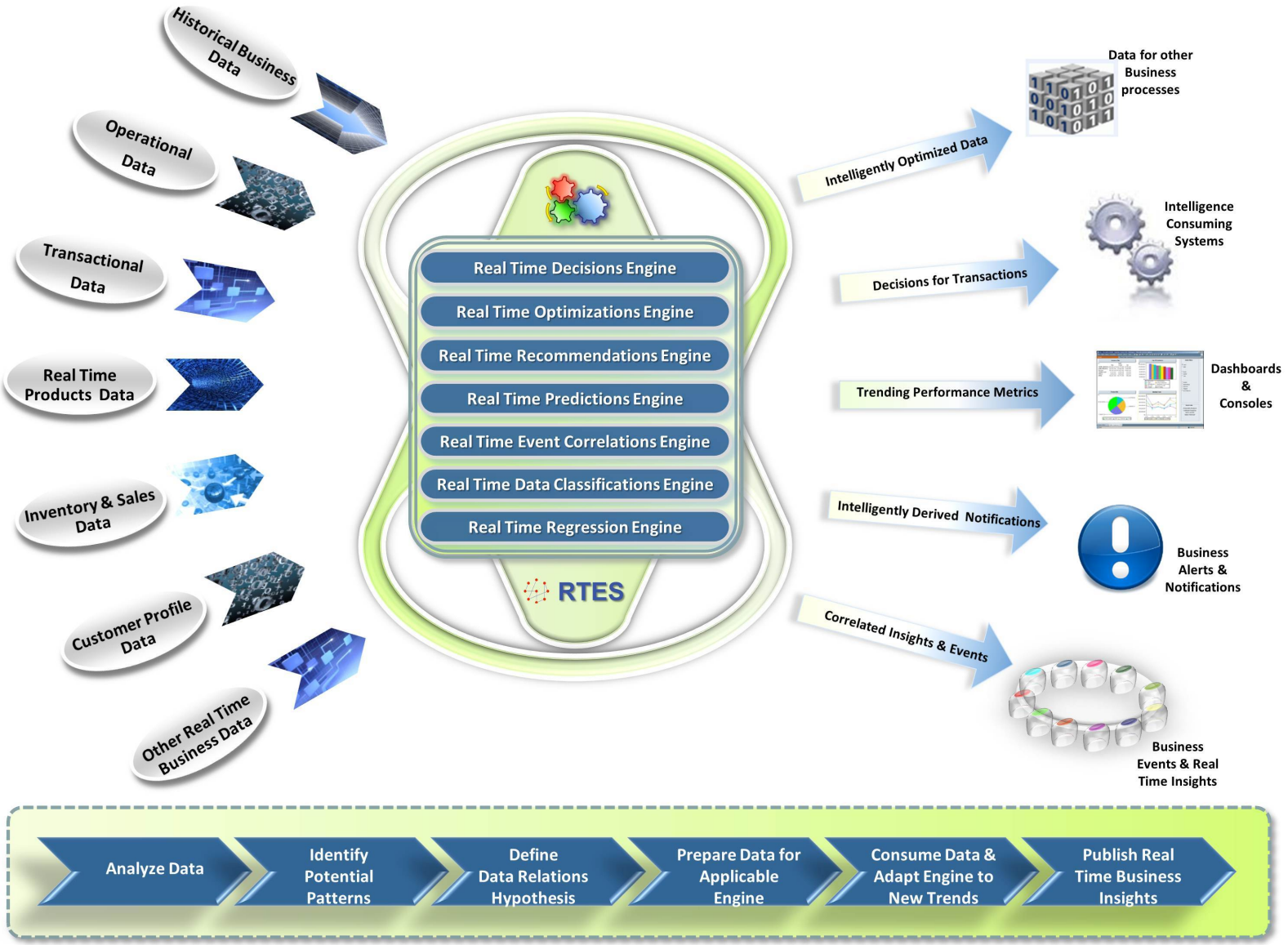
# Modeling Process

- Define Goals
- Gather Data
- Decide Model
- Prepare Data
- Variable Selection and Elimination
- Build Candidate Model
- Finalize Model
- Implementation and Monitoring

# Difficulties in Model Building Process

- Lack of clarity in problem definition
- Using data too old or otherwise not relevant
- using too simple/complex model
- Not considering outliers
- Relying only on linear variable selection or compress techniques
- Going deep on single specialized technique
- Not rebuilding final model optimally using all appropriate data
- Errors in implementation process

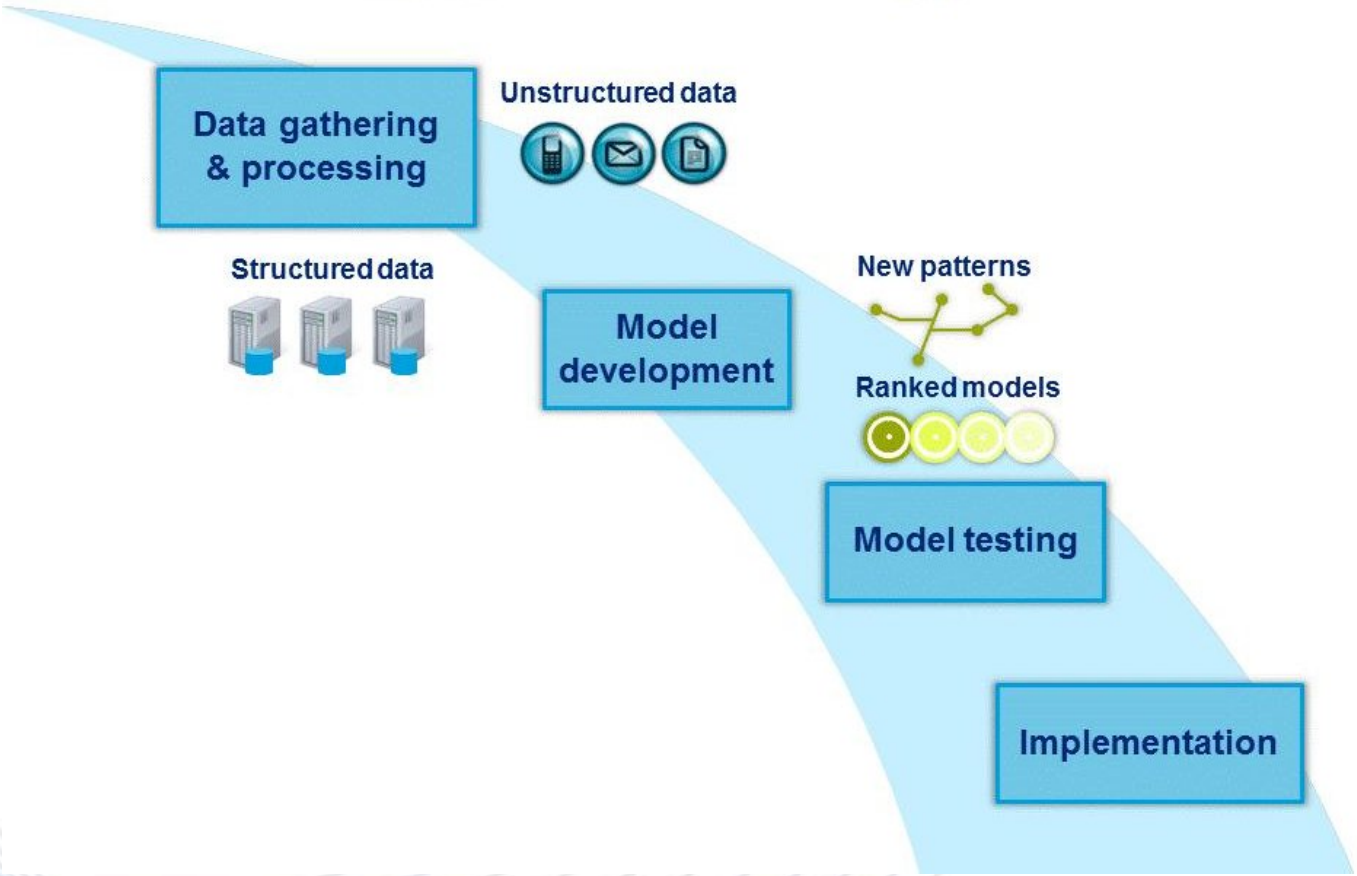
# Topics


Historical Business Data
Operational Data
Transactional Data
Real Time Products Data
Inventory & Sales Data
Customer Profile Data
Other Real Time Business Data
Real Time Decisions Engine
Real Time Optimizations Engine
Real Time Recommendations Engine
Real Time Predictions Engine
Real Time Event Correlations Engine
Real Time Data Classifications Engine
Real Time Regression Engine
RTES
Intelligently Optimized Data
Data for other Business processes
Decisions for Transactions
Intelligence Consuming Systems
Trending Performance Metrics
Dashboards & Consoles
Intelligently Derived Notifications
Business Alerts & Notifications
Correlated Insights & Events
Business Events & Real Time Insights
Analyze Data
Identify Potential Patterns
Define Data Relations Hypothesis
Prepare Data for Applicable Engine
Consume Data & Adapt Engine to New Trends
Publish Real Time Business Insights

# Typical Data Sources

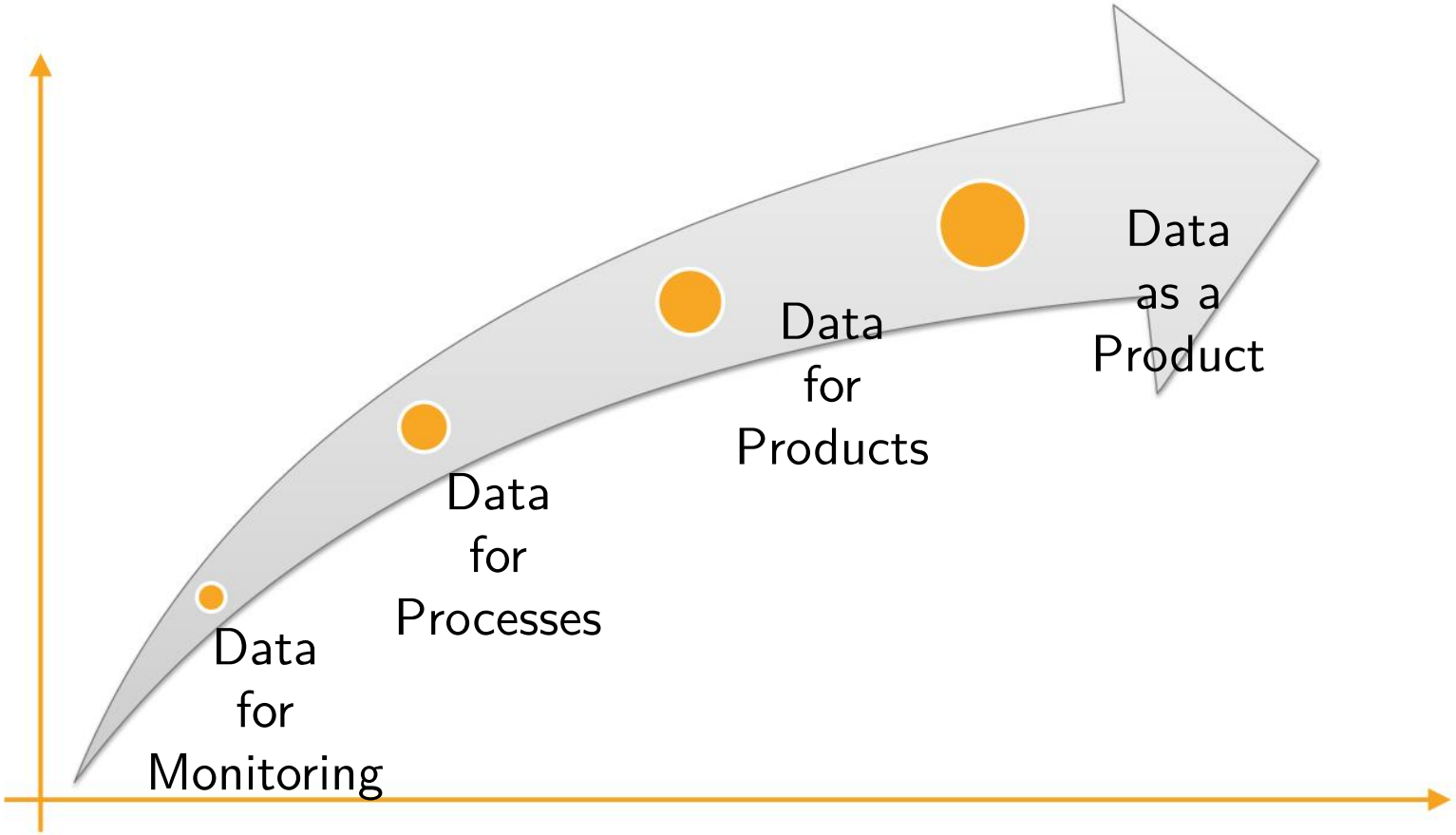
| Classical Systems | New Systems | | |
|---|---|---|---|
| Structure Data<br>Transactions<br>system<br>(ERP, SCM, CRM) | Semi-structured text (e.g. XML) or unstructured text (*.txt)<br>Sensor data<br>Log data<br>M2M | Mobile IT | Social Web |
| - customer data<br>- product data<br>- orders<br>- ... | - production data<br>- weather data<br>- geo data<br>- ... | - RFID data<br>- telecommunications<br>- traffic data<br>- ... | - Twitter<br>- Facebook<br>- LinkedIn<br>- youtube<br>- Blogs<br>- Idots |

# Analytics Process

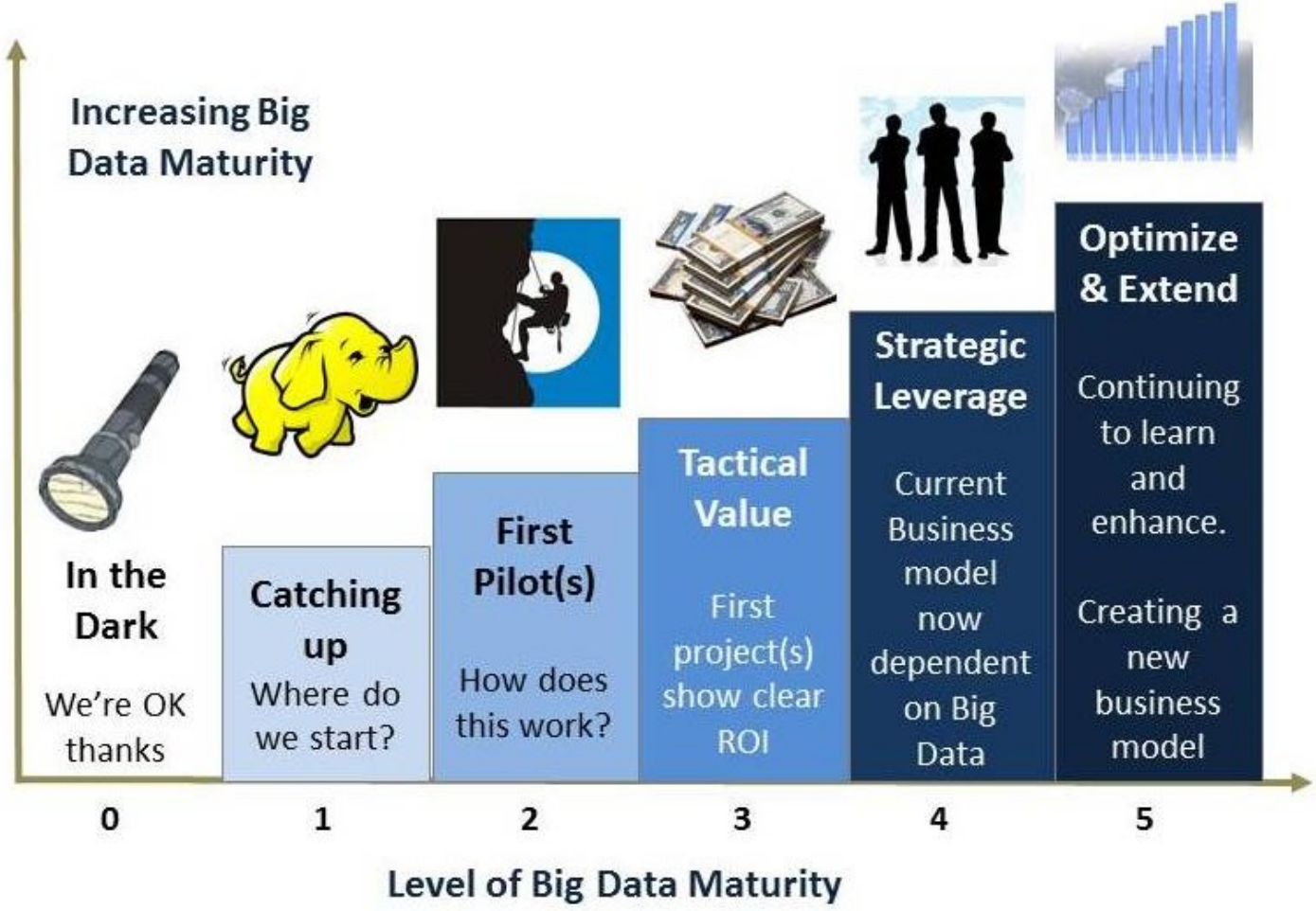



Source: https://sctr7.com

# Data in Industry 4.0 Strategy



# Big Data Maturity Model




Source: Radcliffe Advisory

#BigData_MatMod

# Data Mining - Procedure models

- Knowledge Discovery in Databases (KDD)
  - Process model for discovery from data
  - Procedure for selecting and preparing data
  - Data mining describes analysis methods and is a sub-process step of KDD model
- CRoss-Industry Standard Process for Data Mining (CRISP-DM)
  - Iterative and agile process model for data mining
  - Standard process with six phases for analyzing data sets according to patterns, trends and relationships

$\Rightarrow$ Process models structure the evaluation of Big Data

# Knowledge Discovery in Databases Process

# Knowledge Discovery in Databases (KDD) - General

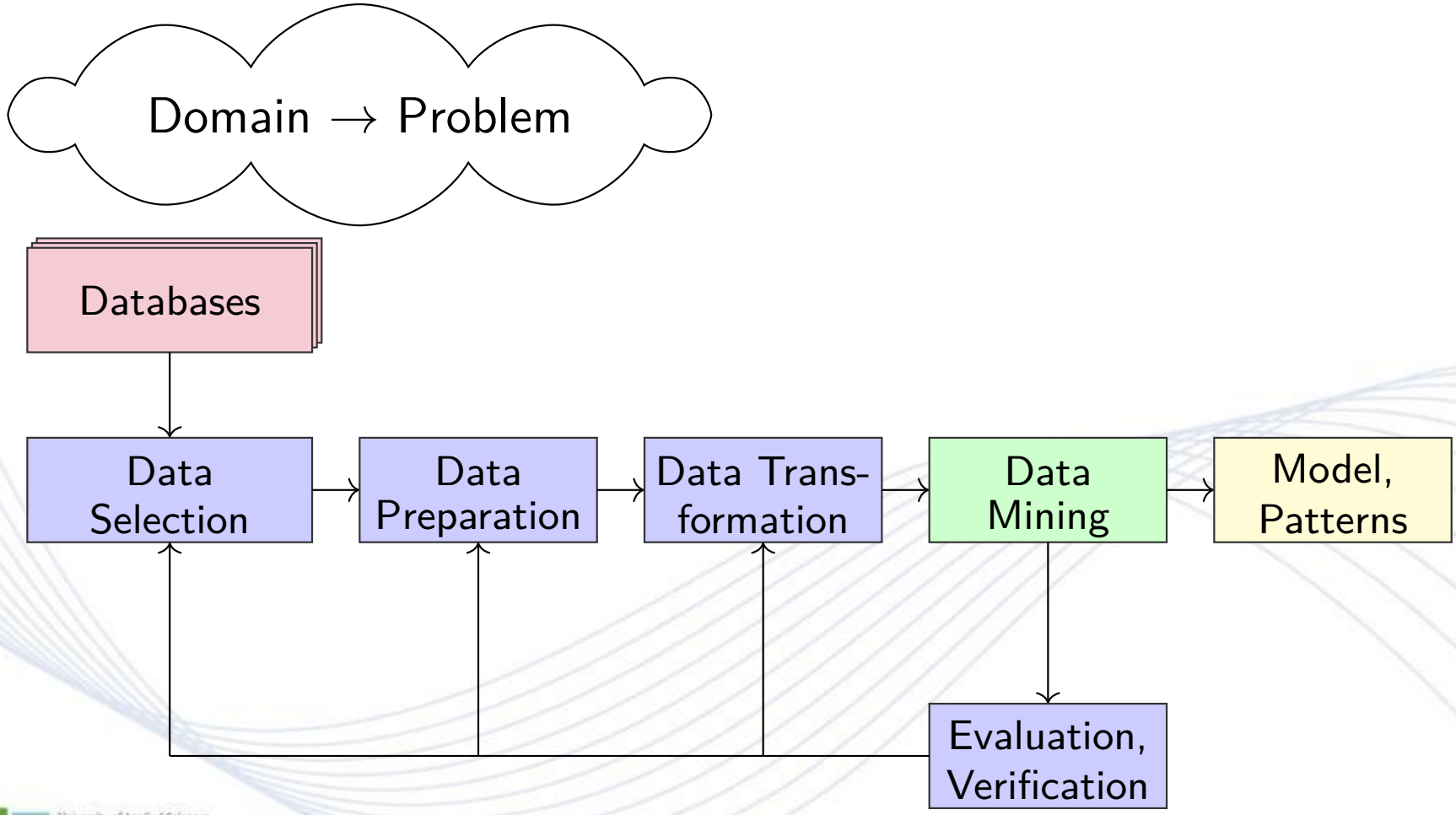
- KDD is an approach to data analysis to discover and make explicit knowledge available in extensive data sets
- Recognition of relationship patterns e.g. regularities or anomalies in large amounts of data
- Derivation of previously unknown correlations that are valid for as large a proportion of the database as possible

# Knowledge Discovery in Databases (KDD) - Process

- Focus on the entire knowledge discovery process
- Comprehensive handling of large amounts of data
- Process includes selection, preparation, specification, analysis and interpretation of data Data Mining is a sub-process

$\Rightarrow$ Comprehensive process for discovering knowledge from databases

# Knowledge Discovery in Databases Process



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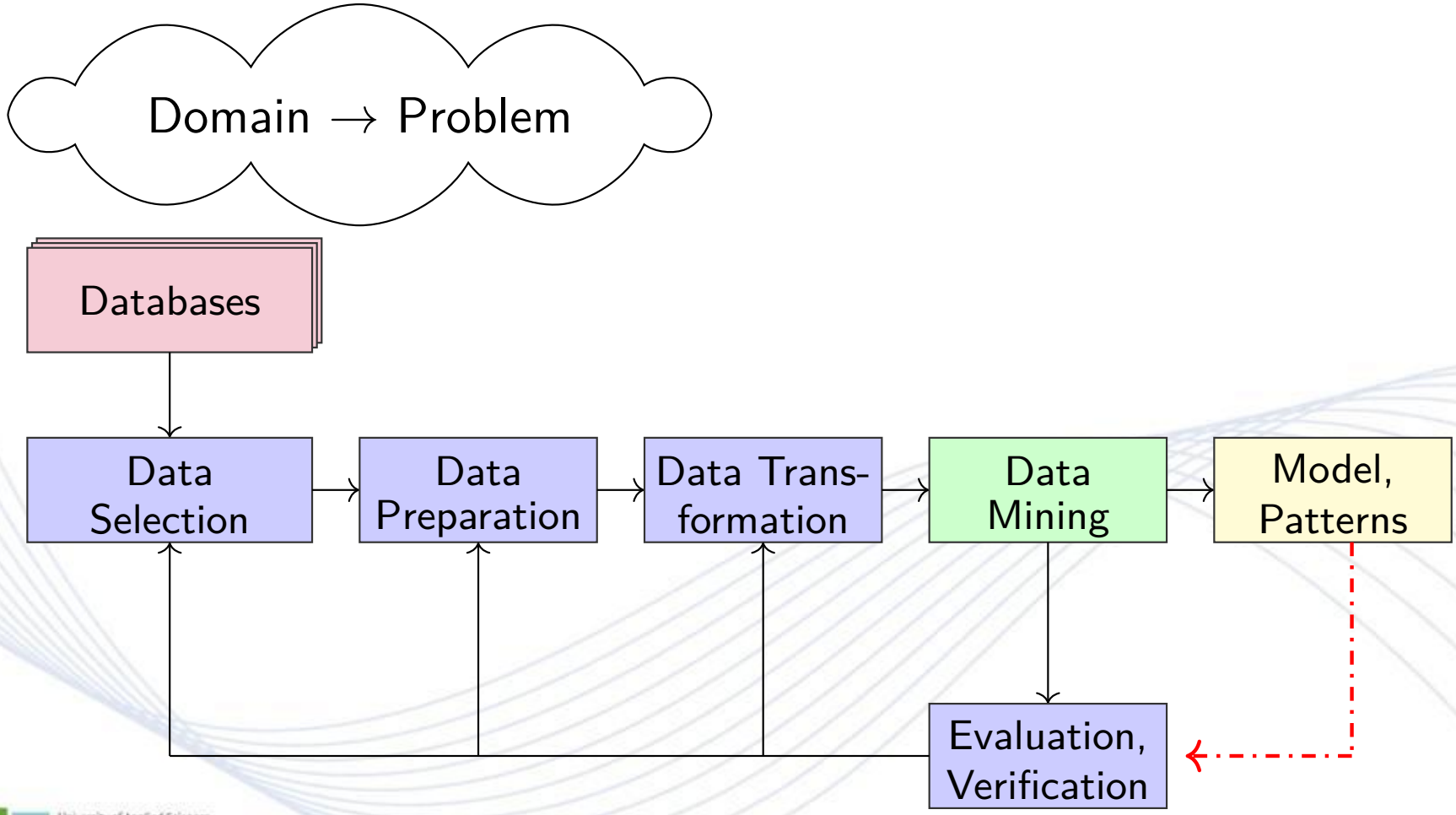
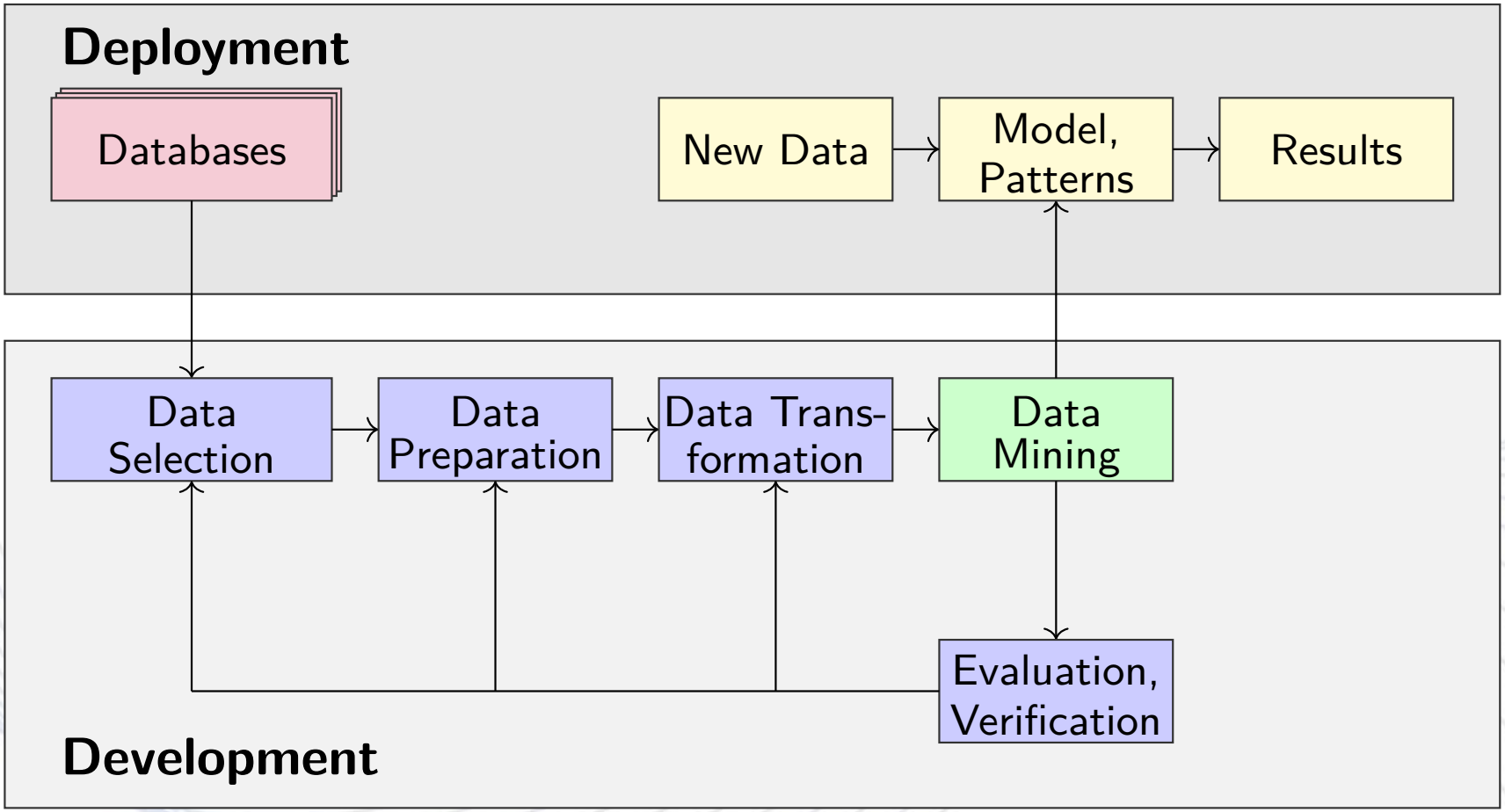


# KDD Process - Machine Learning Pipeline



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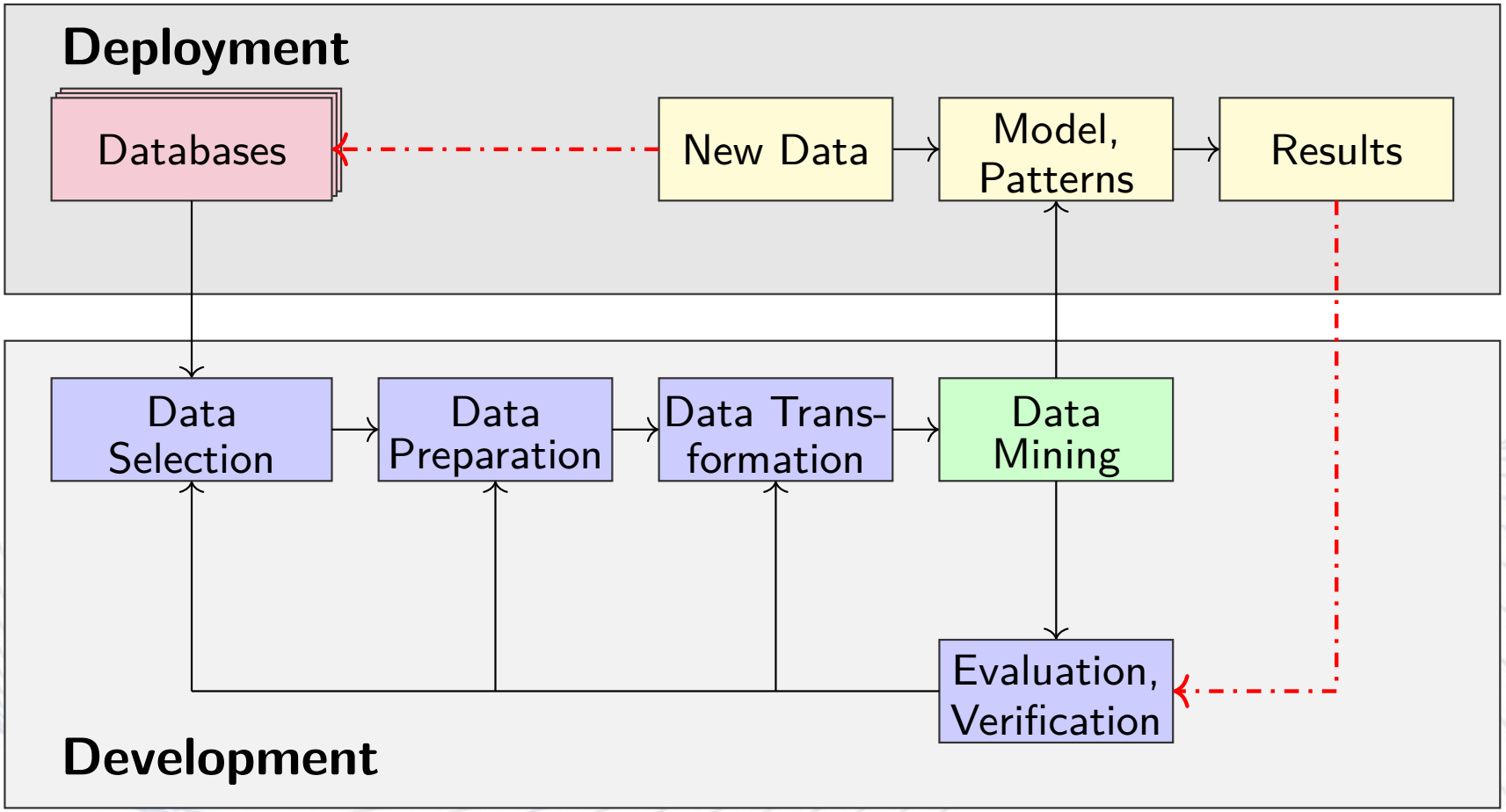

# Knowledge Discovery Databases Process Steps

Learning the application domain:

- relevant prior knowledge and goals of application
  $\Rightarrow$ Creating a target data set: data selection
- Data cleaning and preprocessing: (may take 60% of effort!)
- Data reduction and transformation:
  $\Rightarrow$ Find useful features, dimensionality/variable reduction, representation
- Choosing functions of data mining
  $\Rightarrow$ summarization, classification, regression, association, clustering
- Evaluation/Verification

Iterative execution of the process steps

# KDD Process Steps

Choosing functions of data mining

- Choosing functions of data mining
  $\Rightarrow$ summarization, classification, regression, association, clustering
- Choosing the mining algorithm(s)
- Data mining: search for patterns of interest
- Pattern evaluation and knowledge presentation
  $\Rightarrow$ visualization, transformation, removing redundant patterns, etc.
- Use and integration of discovered knowledge

# Modeling Process

# Modeling Process

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# Database

# Database

I want to know . . .

- What is Data?
- What is information?
- What kind of data do we need?

# Database

I want to know . . .

- What is Data?
- What is information?
- What kind of data do we need?

# Phase: Data selection

- Goals:
  - Determination of the starting position for the process: What data is available? How is the data available?
  - Selection of the relevant data: Which part of the data is relevant? Is further data needed? Are there dependencies between the data?
  - Definition of the task: What should be investigated? (e.g. probability of failure of a machine)
    Which decision should be made? (e.g. premature maintenance of a machine)
- The formulation of the task should be sufficiently general and sufficiently concrete to measure both hypotheses and the success of the process
- Checking the data stock for availability, completeness and admissibility of the analysis

# Database

- Domain Knowledge

  $\Rightarrow$ choose your database:

- Data as much as possible
  - Image: Ok or not Ok
  - Voice: Ok or not Ok
  - Text: Ok or not Ok
  - e.g. (face, name)

# Database

Database is a systematic collection of data;
Computer structures that save, organize, protect, and deliver data

Database Management System (DBMS) is a collection of programs which enables its users to access database, manipulate data, and help in representation of data; also helps control access to the database by various users

# Database

Text Database:

- The simplest form of database
- When data is organized in a text file in rows and columns, it can be used to store, organize, protect, and retrieve data
- For example: name , family name, age , . . .

# Database

Desktop database programs:

- more complex than a text database but intended for a single user
- A Microsoft Excel spreadsheet or Microsoft Access
- 4 major commands:
  - Select,
  - update,
  - insert, and
  - delete

- benefit: speed of changing data, and the ability to store large amounts of data while keeping performance of the system

# Database - Example

Open Datasets:

- MS-COCO
- ImageNet
- VisualQA
- CIFAR-10
- AUVSI: Unmanned Vehicles and Robotics products operating air ground and maritime domain, finding new platforms, new information is added to database daily email

http://www.analyticsvidhya.com/blog/2018/03/comprehensive-collection-deep-learning-datasets/

# Example: Definition of the problem "Inverse Kinematic"

Now we define our problem:

We want to calculate the inverse kinematic a serial robot.

# Example: Definition of the problem "Inverse Kinematic"

Normally, there are serveral algorithms in use:

- Newton-Raphson,
- Newton-Raphson with Jacobian Transpose
- Cyclic Coordinate Descent (CCD),
- Damped Least Squares (Levenberg-Marquardt),
- Broyden-Fletcher-Shanno (BFSD) algorithm
- Lagrange Multiplier
- ...

If you want to use this algorithms, their time consumption is not well-defined.
Therefore, you have to use special hardware in a real-time environment.

# Example: Definition of the problem "Inverse Kinematic"

Now, we use a neural network

- well-defined time consumption
- fast
- easy to implement

Question:

- accuracy?
- reliable?

# Application - Direct and Inverse Kinematics

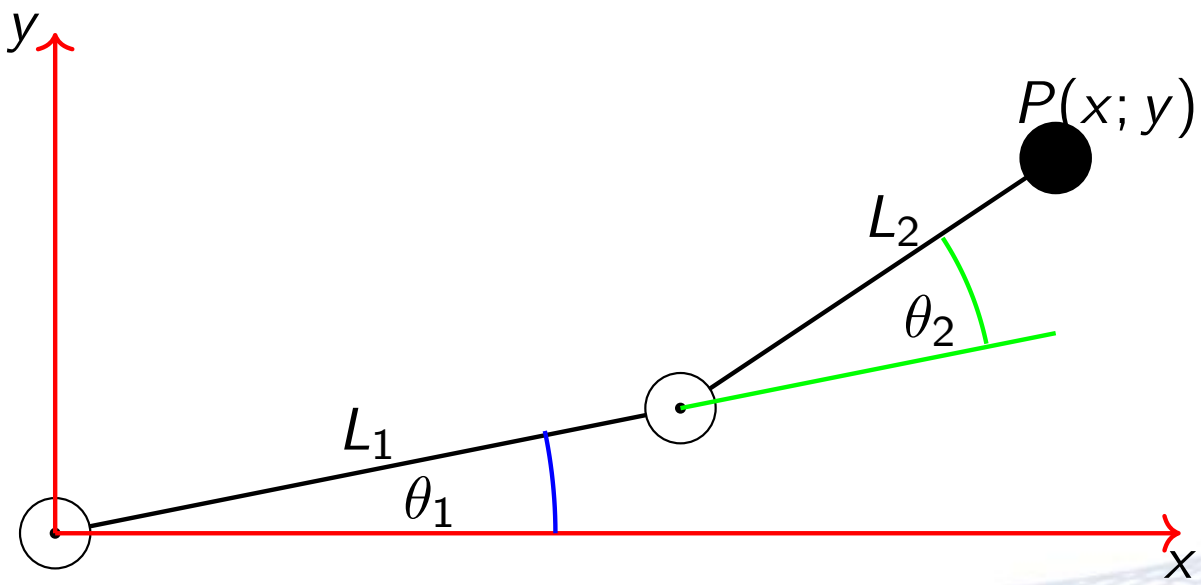
Database application example $\Rightarrow$ end-effector position

$$\text{input: } (\theta_1, \theta_2) \rightarrow \text{output: } \binom{x}{y}$$

$\theta_1, \theta_2$ are joint position

x,y are end-effector positions

# Example: Direct Kinematic of a SCARA




Is $P(x, y)$ the end point of the second arm, so we get for the coordinates of $P$:

$$\begin{aligned} x &= L_1 \cdot \cos(\theta_1) + L_2 \cdot \cos(\theta_1 + \theta_2) \\ y &= L_1 \cdot \sin(\theta_1) + L_2 \cdot \sin(\theta_1 + \theta_2) \end{aligned}$$

# Database Application

Direct Kinematic

$$(\theta_1, \theta_2) \quad \rightarrow \quad (x, y) \quad \Rightarrow \quad \text{data set} \quad (\theta_1, \theta_2, x, y)$$

How much data?

As much as possible?

# Database Application

Better: **a plan!**

$\rightarrow$ covering the working space

$\rightarrow$ considering a tolerance $\varepsilon_1$

$$\left\| \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} - \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right\| < \varepsilon_1$$

as little as possible, but enough

$\Rightarrow$ creating a database based on domain knowledge

# Selection

# Data selection

- Goals:
  - Determination of the starting position for the process: What data is available? How is the data available?
  - Selection of the relevant data: Which part of the data is relevant? Is further data needed? Are there dependencies between the data?
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# Selection

Data mapping: The process of defining how individual fields are mapped, modified, joined, filtered, aggregated etc. to produce the final desired output. Developers or technical data analysts traditionally perform data mapping since they work in the specific technologies to define the transformation rules (e.g. visual ETL tools, transformation languages).

# Selection

- Code generation: the process of generating executable code (e.g. SQL, Python, R, or other executable instructions) that will transform the data based on the desired and defined data mapping rules. Typically, the data transformation technologies generate this code based on the definitions or metadata defined by the developers.

- Code execution: The step whereby the generated code is executed against the data to create the desired output. The executed code may be tightly integrated into the transformation tool, or it may require separate steps by the developer to manually execute the generated code.

# Selection

- Data review: The final step in the process, which focuses on ensuring the output data meets the transformation requirements. It is typically the business user or final end-user of the data that performs this step. Any anomalies or errors in the data that are found and communicated back to the developer or data analyst as new requirements to be implemented in the transformation process.

# Example Direct and Inverse Kinematic

As an example in robotics some data may have different unit, which need to be unified:

- Angle, (Degree, Radian, ...)
- Rotation speed (Revolution per minute, Radian per second, ...)
- Operation modes (mode A, mode B)
- Also, some data may not be relative to the knowledge we are interest to extract.
- For example, if we are investigating about angular speed, the data about the temperature may not be applicable.

# Selection

Inverse Kinematic

- Not easy
- Not unique answer

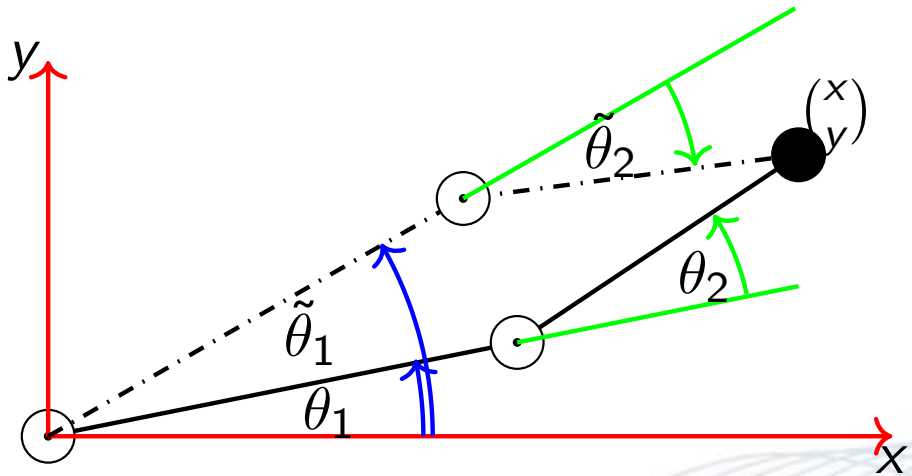
but

Direct Kinematic

easy to calculate $\rightarrow$ $F(\theta_1, \theta_2) = \binom{x}{y}$

# Selection - Calculate the data

calculate all values using $F$ and put them in a database




$$F(\theta_1, \theta_2) = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = F(\tilde{\theta}_1, \tilde{\theta}_2)$$

# Selection - Calculate the data

calculate all values using $F$ and put them in a database, but

$$\tilde{\theta}_2 = -\theta_2$$

choose, e.g.:

$$\theta_2 > 0$$

We can not use all combinations!
It depends on the application!!

# Selection

Unique solution is needed!

restriction is needed! $\leftarrow$ we need domain knowledge!!

# Transformation

- $\theta$'s unit is radiant or degree?
- Definition set:
  - $\theta \in [0; \pi]$?

    or
  - $\theta \in [-75°; 280°]$?

    or
  - $\theta \in [0; 1]$ (that's a standard!)?

    or
  - $\theta \in [-1; 1]$ (that's a standard, too!)?

$\Rightarrow$ depending on the algorithm $\theta$ will be chosen

# Data Preparation

# Data Preparation

How does the algorithm need the input data?

- list
- list of lists
- arrays

Is there a difference between

- just input
- input and training

# Data preparation

- Goal:
  Quality improvement of the selected data regarding completeness and consistency
- Differentiation in data organization and presentation based on several data sources, transfer to a uniform database; e.g. merging of data from the machine and from work preparation
- Adjustment approaches to the data:
  - Enrichment: Subsequent collection of missing and incorrect values is very costly
  - Reduction: Removal of data records with missing and incorrect values can lead to loss of relationship patterns
  - Integration: Replacement of missing and incorrect characteristics or addition of further characteristics leads to higher complexity
- Feasibility of the analysis depends on complexity

# Data Preparation

How does the algorithm deal with anomalies?

- Missing values?
- Checking values? plausible?
- outliers?

# Data Preparation

In a knowledge extraction process, all of the process is relied on the data source. Good quality input data leads to good quality results. In KDD process:

Garbage in - Garbage out.

The raw recorded data may have different resolution, bias, range, validity, noise, or etc. Always there is a chance that a datum is not recorded properly which may ruin the knowledge extraction process.

# Data Preparation - Example

As an example, in a robotic system some gathered data may not be use and should be removed or replaced:

- Disconnected/broken sensor
- Dirty sensor
- Too much noise
- Not calibrated
- Data is recorded in a undesired time span

# Data Transformation

# Data Transformation

- Goal:
  Definition of the objective and the analysis procedure

# Data Transformation

- Possible objectives of KDD:
  - Formation of groups: summarize similar data sets, derive knowledge about the similarity and diversity of data sets in the database, e.g. market segmentation
  - Explanation of groups: Determine the functional relationship between qualitative grouping characteristics and quantitative or qualitative characteristics, to justify the relationship of groups, e.g. risk assessment of clients
  - Explanation of correlations: Determine knowledge from determination of a relationship between a dependent quantitative characteristic and one or more independent quantitative or qualitative characteristics of the data sets, e.g. the impact measurement of marketing measures
  - Description of correlations: Determine associations between the characteristics of the data sets, e.g. sequence analysis determines the interrelationships of the temporal purchase behavior of customers

# Data Transformation

- The type and sequence of the analysis are determined by determining the analysis procedure

# Data Transformation

Split the data:

$$\text{Data} \to \begin{cases} \text{Training Data} \\ \text{Test Data} \end{cases}$$

How?

# Data Transformation

Split the data:

How?

The normal proportion is 80% training data and 20% test data.

- randomly?
- first 80%
- every $5^{\text{th}}$ element is a test data
- ...

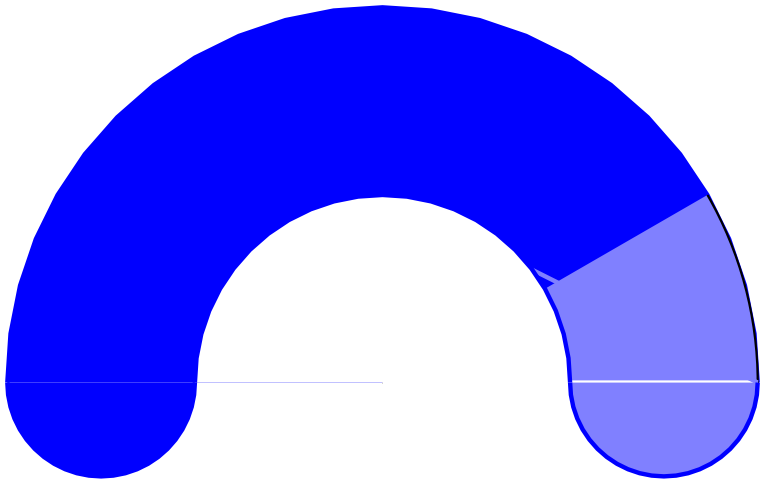
# Data Transformation

Normally choosing data isn't optimal!

Use different strategies.

# Data Transformation - Example

Covering the working space?



Perhaps, you get 80% left side of data

Is some of data missing?

$\Rightarrow$ Domain knowledge is useful

# Data Transformation

A large part of variables in a robotic system are dynamic values, and their values change in during the operation. Therefore the true knowledge can be discovered, only when the values of a parameter is known in time domain.

Normally are all recorded data in a database for a robotic system is a time-stamped observation, described by a set of feature-value pairs.

The time value is given on the time scale in a given resolution, such as millisecond, days, months, or years.

# Data Transformation - Example Direct and Inverse Kinematic

The data transformation is critical to detect important events in robotic systems such as:

- Predicting events and behaviors
- Change points in time domain
- Parameters associated with time, like: velocity and acceleration etc.
- Aging process
- Trajectory

# Data Mining

# Data Mining

- Goal:
  Determine and map relationship patterns in the database
- Approach to data analysis in order to determine relationship patterns in a database and to map them through logical or mathematical descriptions
- KDD not only determines and maps relationship patterns, but also a process with data mining as a sub-step
- Data Mining is an analytical approach to test hypotheses by means of questions such as "How many units of the product group backpacks were sold in all stores using the standard shipping method?"
- Possible result of the analysis in the form of a model: "The sale of goods of the departmental group backpacks in the North branch was carried out via the standard shipping method in 95% of the cases."

# Data Mining

Data mining is a process of finding previously unknown patterns in databases to build predictive models

Learning Algorithms:

- Supervised Learning
- Unsupervised Learning
- Reinforced Learning

# Supervised Learning

Not only the algorithm has the input, but also the desired output
complete and labelled data are given to the algorithm
A teacher guides the system to learn the pattern
The algorithm should predict and put labels to the unlabelled data
It has feedback mechanism

Supervised Algorithms:

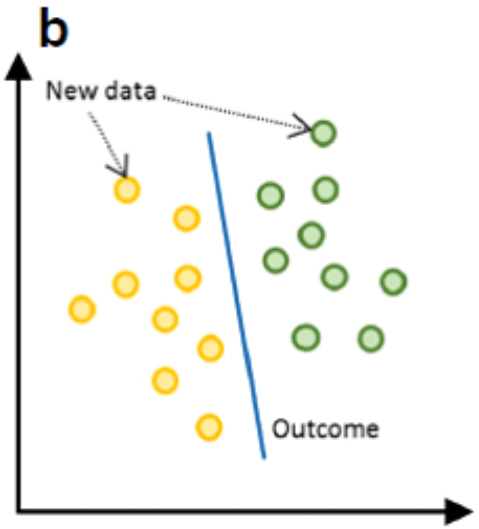
- Decision Tree
- Random Forest
- Support Vector Machine
- Artificial Neural Networks
- K Nearest Neighbors

# Supervised Learning Types

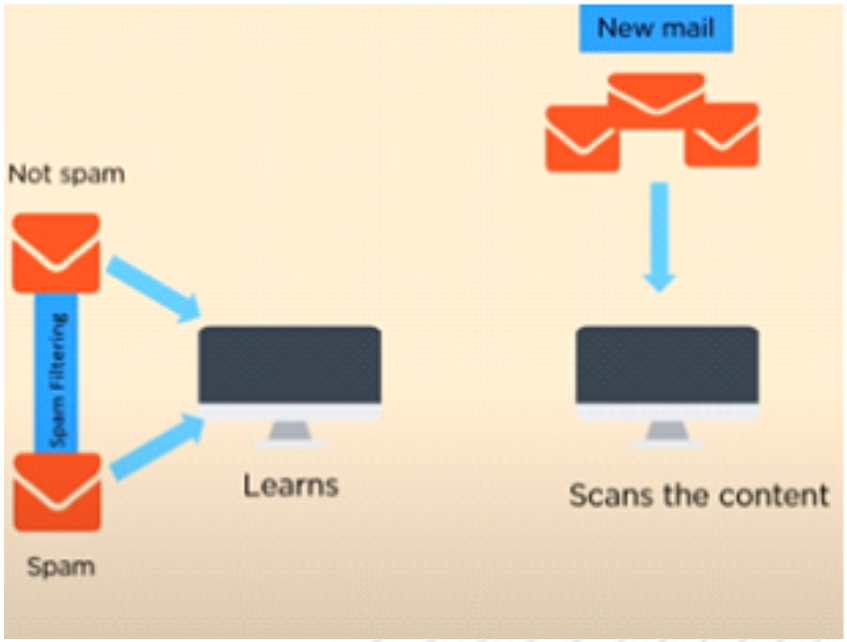
- Classification
- Regression

Classification: When the output variable is categorical e.g. 2 or more classes (yes/no, true/false), we make use of classification

Classification Examples:



# Supervised Learning Types

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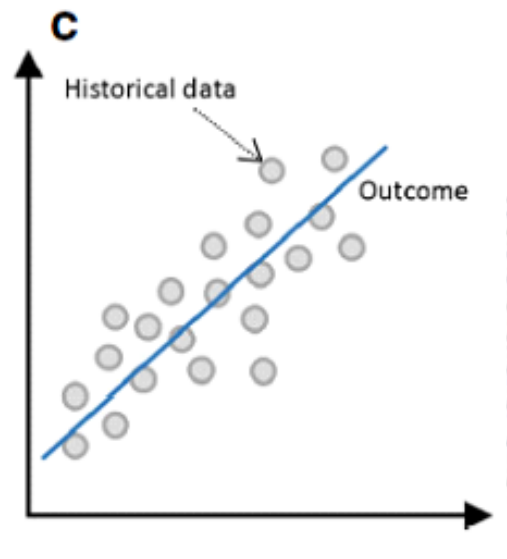
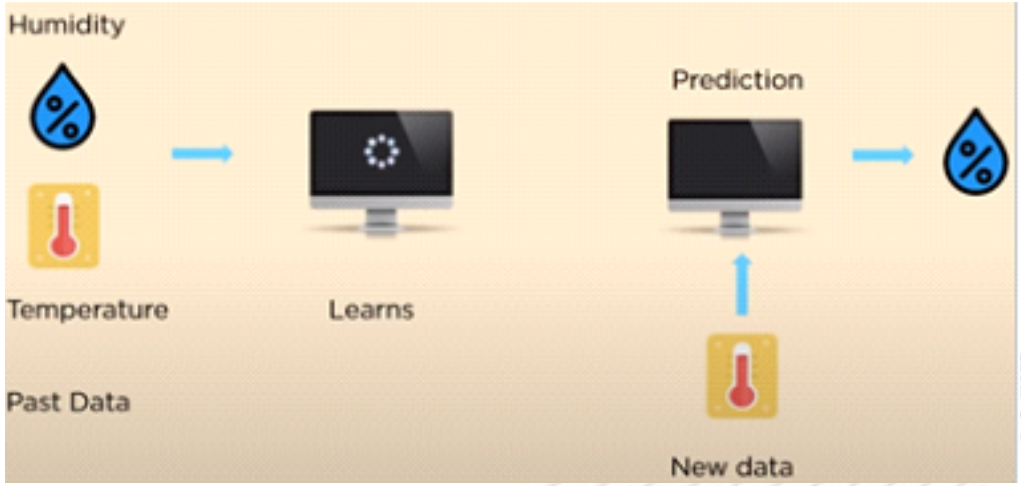
# Supervised Learning Types

Regression: Relationship between two or more variables where a change in one variable is associated with a change in other variable there are some examples of Regression:



# Supervised Learning Types

Weather Prediction

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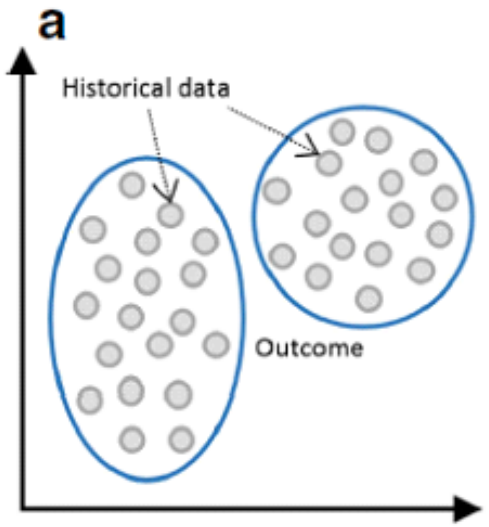
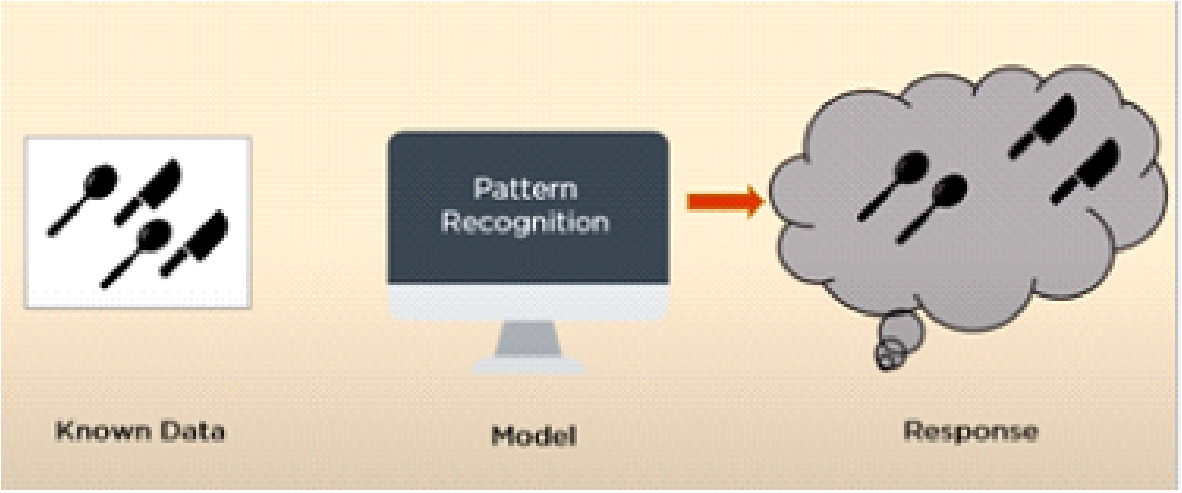
# Unsupervised Learning

Unsupervised Learning: The desired output is not available, the algorithm is trained using data that is unlabelled. It does not have feedback mechanism.

Unsupervised Learning Example:



# Unsupervised Learning

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Unsupervised Algorithms:

- K Means
- Mean Shift Clustering

# Unsupervised Learning Types

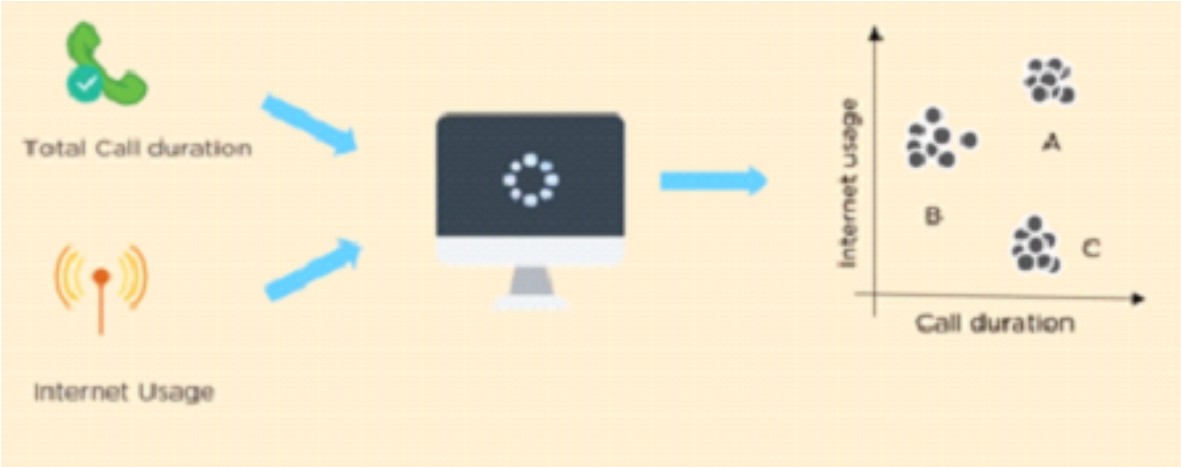
- Clustering
- Association

**Clustering**: The method of dividing the objects into clusters which are similar between them and are dissimilar to the objects belonging to another cluster

# Unsupervised Learning Types

There are examples of Clustering:

An example of Clustering: Suppose a telecome company wants to reduce its customer churn rate by providing personalized call and data plans:

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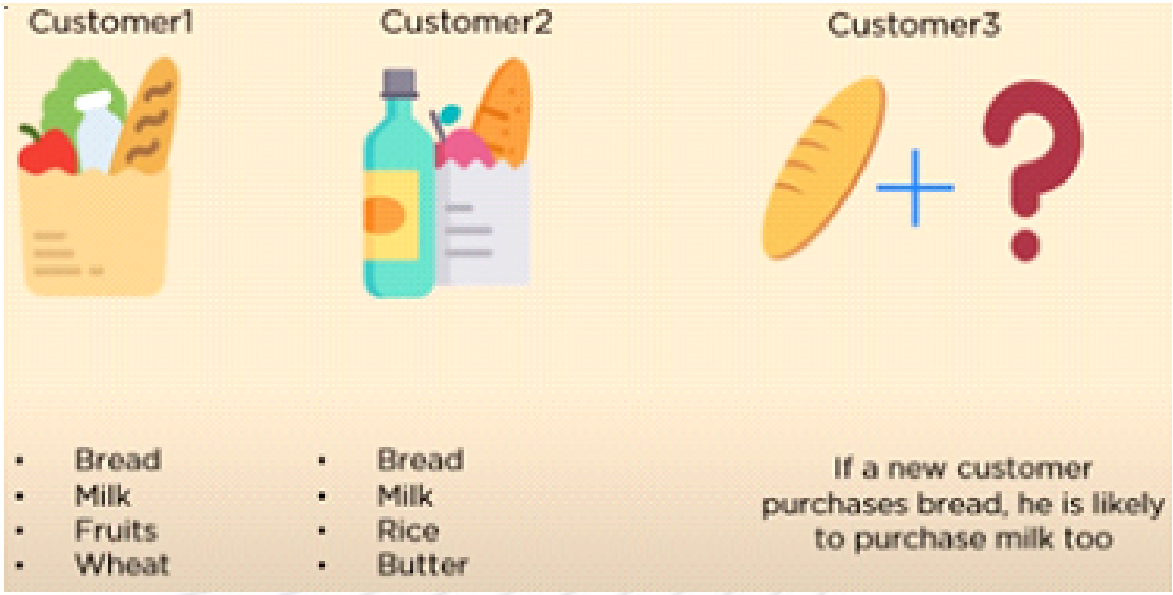
# Unsupervised Learning Types

**Association**: Discovering the probability of the co-occurrence of items in a collection

There are some examples of Association:

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# Reinforced Learning

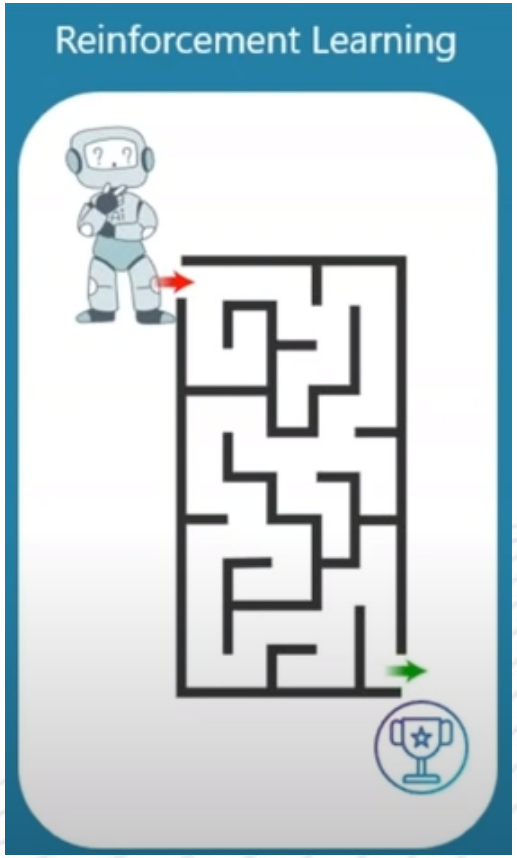
Reward and punishment based learning. For certain ways achieving a target, we reward the algorithm and for other ways of achieving we punish the system.

We encourage the algorithm to find the solution or achieve the target in such a way that is most rewarding and least punishing

Example: Reaching to a toy by a child or robot

# Reinforced Learning

Example: Reaching to a toy by a child or robot




Source: my own

# Reinforced Learning

Reinforcement Learning Algorithms:

- Policy Gradient
- AlphaZero

# Data mining applications

- Data mining is a process of finding previously unknown patterns in databases to build predictive models.
- Data mining Applications:
- Robot Control
  - Routing, Localization in WSN $\rightarrow$ Genetic Algorithm
  - Control a robotic arm and gripper $\rightarrow$ Support Vector Machine
  - Behavioral model for prevention of collision situation
  - Constructing a map of unknown environment by autonomous robots $\rightarrow$ K-means

# Data mining applications - Robot Vision

- Being aware of environment by rescue autonomous robots which are equipped with vision system.
- Making decision based on visual input (image and video)
- Autonomous Underwater Robots imagery of biological species, physical formations, and object detection
- Robot Behavior
- classify robot actions for imitation learning by hybrid neural data mining technique
- Robot Behavior
  - Improving robots' behavior through web mining, social media, and experienced patterns as a knowledge source alongside with contextual information to discover users' needs

# Data mining application - Inverse Kinematics

We use a simple neural network with one hidden layer and 50 neurons.

# Model Patterns

# Model Patterns

Is the most important phase of a knowledge discovery process
This model can be

- Detecting temporal and sequential patterns
- Expressed as fuzzy relationships between events in consecutive periods

# Model Patterns

The model extraction can be done within a supervised or unsupervised KDD process.

- In supervised process: the model of the system is known
- In unsupervised process, the mode is not well known, and will be detected using the data (e.g. the clustering the vibration patterns of the system which may or may not lead to a system fault.)

# Model Patterns - Example

Example of application of discovered models and patterns:

- Providing a detailed model which describes the behavior of a complicated robotic system that can help us to control the system in a more efficient way.
- Fault detection and trouble shooting..
- Detecting the suspicious behavior change of a system, which can be an early alarm for necessity of maintenance.

# Model Patterns - Inverse Kinematics

We implement a function with the neural network in the robot's control system.

The best way:

In our development environment, we save the neural network using a standard data exchange format, e.g. onnx or nnef.
We implement a function in the robot's control system which read in the neural network.

# Evaluation, Verification

# Definition

**Evaluation**:

Evaluation refers to the systematic analysis and assessment of measures and processes.

**Verification**:

Verification is the proof that a presumed or asserted fact is true.

# Interpretation

- Goal:
  Derivation of explicit knowledge from identified relationship patterns
- Evaluation of analysis results
- Explicit knowledge about contextual references is generated, examined by the hypothesis generation in the KDD process
- Determined correlations can be used by the user for improvements or changes
- Example:
  - Context: "When a customer buys paint, in 90% of cases he also buys paint brushes."
  - Future measure: Positioning brushes and paints together in a way that is tangible for the customer to increase sales

# Evaluation, Verification

The last step in a KDD process is to verify and evaluate the results.

A common approach for this process is cross check.

In this approach a part of dataset will be reserved and not used in the knowledge extraction process. After the model is extracted, we will check whether the model is also valid for the reserved data or not.

# Evaluation, Verification - Example Direct and Inverse Kinematic

In a robotic system, besides the cross-check also usually we can validate the result by comparing it with the known fundamental physics equations. For example the extracted knowledge normally should never contradict with the mass conservation, energy conservation or Newton's laws of motion.

# Review of Results

Review of results throughout the KDD process

- The process should be automated if possible, but: Change the situation, e.g. COVID-19
- Change of the process, e.g. defects or new sensors
- Improved algorithms, e.g. Tensorflow 7.0
- Improved data management, e.g. doubling the amount of data
- ...

$\Rightarrow$ KDD must be questioned again and again
$\Rightarrow$ KDD must be set up again if necessary.